

Beauty production studies via quarkonia measurements

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Abstract

Beauty production in proton–proton collisions provides a test of perturbative QCD calculations, constrains beauty-quark hadronization, and establishes a reference for measurements in heavy-ion collisions. These proceedings report preliminary ALICE measurements of beauty production at midrapidity in pp collisions at $\sqrt{s} = 13.6$ TeV using two complementary approaches. In the first approach, the contribution from beauty-hadron decays to J/ψ and $\psi(2S)$ production is determined through measurements of their non-prompt fractions, including the first preliminary ALICE measurement of $\psi(2S)$. In the second approach, $B^\pm$ mesons are exclusively reconstructed through the decay $B^\pm \rightarrow J/\psi K^\pm$, with $J/\psi \rightarrow e^+e^-$. The resulting p_T -differential $B^\pm$ production cross section extends the kinematic reach of fully reconstructed $B^\pm$ measurements at the LHC towards lower p_T at midrapidity, reaching a previously unexplored kinematic region. It is broadly described by state-of-the-art perturbative-QCD calculations within uncertainties.

1. Introduction

Beauty quarks have a large mass and are produced in hard partonic scatterings with a large momentum transfer. They are therefore produced predominantly in the initial stage of the collision, and their production cross section can be calculated using perturbative QCD. Measurements of beauty production in pp collisions serve three main purposes: they test perturbative QCD calculations, provide information about beauty-quark hadronization through comparisons among different beauty-hadron species, and establish the reference needed to quantify modifications of beauty production in Pb–Pb collisions.

The preliminary results presented in these proceedings were obtained with the ALICE detector at midrapidity in pp collisions at $\sqrt{s} = 13.6$ TeV. Beauty production is studied by measuring charmonia, bound states of a charm quark and its antiquark, produced in beauty-hadron decays. The charmonium states considered here are the J/ψ and $\psi(2S)$ mesons. The J/ψ mesons are reconstructed through the dielectron decay channel, while the $\psi(2S)$ mesons are reconstructed through the decay $\psi(2S) \rightarrow J/\psi \pi^+\pi^-$, with $J/\psi \rightarrow e^+e^-$. The two approaches provide complementary information about beauty production. In the first approach, non-prompt charmonia, originating from beauty-hadron decays, are measured to obtain information inclusive over beauty-hadron species. In the second approach, the exclusive reconstruction of individual beauty hadrons in decay channels containing a charmonium state allows the production of a specific species to be measured.

The exclusive reconstruction of beauty hadrons in ALICE was enabled by the comprehensive detector upgrade prior to Run 3. The new Inner Tracking System provides improved tracking, impact-parameter resolution, and secondary-vertex reconstruction close to the interaction point, which are essential for identifying the displaced decay topology of beauty hadrons [1]. The upgraded Time Projection Chamber supports continuous readout at higher interaction rates [2]. Together with the online selection of signal candidates, these improvements provide both the spatial resolution and the large data samples required to reconstruct exclusive beauty-hadron decays despite their small branching fractions and limited reconstruction efficiencies. The exclusive measurement of $B^\pm$ mesons presented here accesses a previously unexplored low- p_T region for fully reconstructed $B^\pm$ mesons at midrapidity at the LHC.

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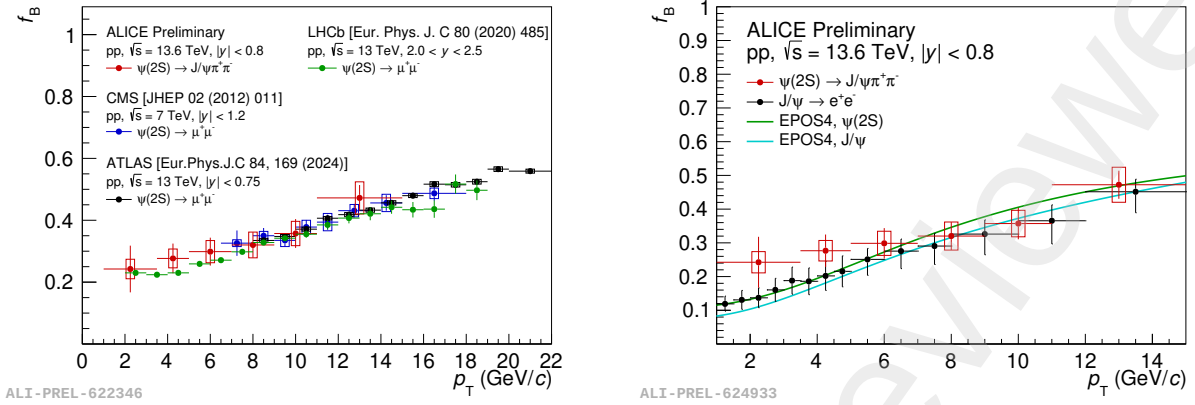


Figure 1: Non-prompt charmonium fraction, f_B , as a function of p_T in pp collisions. Left: non-prompt fraction of $\psi(2S)$ mesons as a function of p_T measured by ALICE in pp collisions at $\sqrt{s} = 13.6$ TeV and $|y| < 0.8$, compared with measurements from CMS at $\sqrt{s} = 7$ TeV and $|y| < 1.2$ [4], ATLAS at $\sqrt{s} = 13$ TeV and $|y| < 0.75$ [5], and LHCb at $\sqrt{s} = 13$ TeV and $2 < y < 2.5$ [6]. The ALICE measurement uses the decay channel $\psi(2S) \rightarrow J/\psi \pi^+ \pi^-$, with $J/\psi \rightarrow e^+ e^-$, while the other measurements reconstruct $\psi(2S) \rightarrow \mu^+ \mu^-$. Right: non-prompt fractions of J/ψ and $\psi(2S)$ mesons measured by ALICE in pp collisions at $\sqrt{s} = 13.6$ TeV and $|y| < 0.8$, compared with EPOS4 calculations [7, 8].

2. Beauty production through non-prompt charmonia

The J/ψ and $\psi(2S)$ mesons are classified as prompt or non-prompt according to their production origin. Prompt charmonia are produced at the primary collision vertex, either directly from a $c\bar{c}$ pair or through decays of higher charmonium states, whereas non-prompt charmonia originate from weak decays of beauty hadrons. Only the non-prompt contribution therefore provides access to beauty production and must be separated from the prompt contribution. This separation is possible because beauty hadrons travel a measurable distance before decaying, typically of the order of a few hundred micrometers; for example, $c\tau_{B^+} \approx 491.1 \mu\text{m}$ [3]. The resulting difference in decay length can be used to distinguish prompt from non-prompt charmonia.

To quantify the contribution from beauty-hadron decays to charmonium production, the prompt and non-prompt charmonium yields are extracted through a simultaneous maximum-likelihood fit to the two-dimensional distribution of invariant mass and pseudoproper decay length. The non-prompt fraction, f_B , is then defined as the non-prompt yield divided by the inclusive yield:

$$f_B = \frac{N_{\text{non-prompt}}}{N_{\text{prompt}} + N_{\text{non-prompt}}}, \quad (1)$$

where N_{prompt} and $N_{\text{non-prompt}}$ are the extracted prompt and non-prompt charmonium yields, respectively.

With the larger Run 3 data sample, the ALICE measurement of non-prompt charmonium production has been extended from the J/ψ to the $\psi(2S)$. The left panel of Figure 1 presents the first preliminary ALICE measurement of the non-prompt fraction, f_B , of $\psi(2S)$ mesons, compared with measurements from CMS [4], ATLAS [5], and LHCb [6]. Despite the different collision energies, rapidity intervals, and decay channels, the measurements show a similar increase of f_B with p_T . This indicates that beauty-hadron decays contribute an increasingly large fraction of the inclusive $\psi(2S)$ yield toward higher p_T .

The right panel compares the ALICE measurements of the non-prompt fractions of J/ψ and $\psi(2S)$ mesons with each other and with EPOS4 calculations [7, 8]. At low p_T , the measurements show a hint that the non-prompt fraction may be larger for the $\psi(2S)$ than for the J/ψ , which could indicate a larger relative contribution from beauty-hadron decays to $\psi(2S)$ production in this region. EPOS4 provides a reasonable description of the observed p_T dependence.

3. Exclusive reconstruction of $B^\pm$ mesons

The exclusive reconstruction of individual beauty-hadron species provides information complementary to that obtained from non-prompt charmonia by giving direct access to the parent-hadron production kinematics. In this analysis, the $B^\pm$ mesons are reconstructed through the decay $B^\pm \rightarrow J/\psi K^\pm$, with $J/\psi \rightarrow e^+ e^-$.

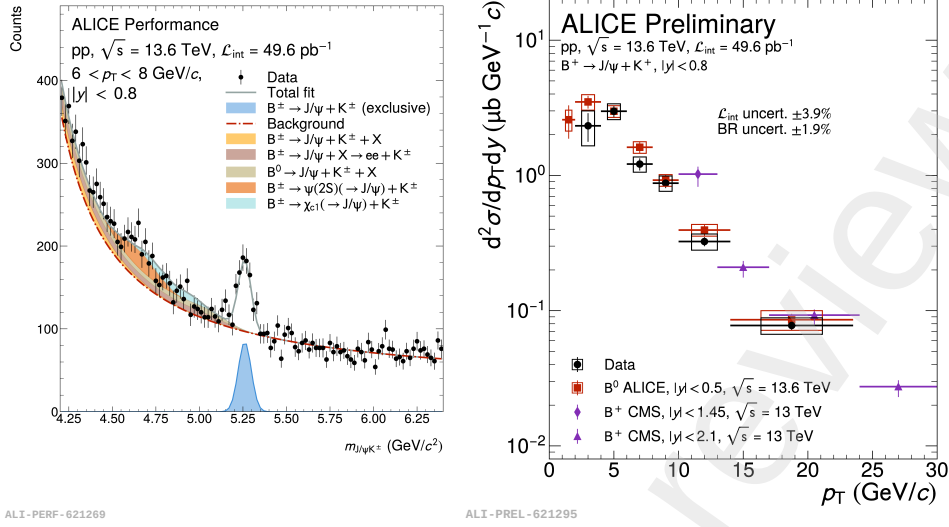


Figure 2: Left: Invariant-mass distribution of $B^\pm$ candidates in pp collisions at $\sqrt{s} = 13.6$ TeV in the interval $6 < p_T < 8$ GeV/c and $|y| < 0.8$. The signal is modeled with a Gaussian, the combinatorial background with a parameterized function, and the correlated-background contributions using templates from Monte Carlo simulations. The individual correlated-background sources are shown separately, as indicated in the legend. Right: Charge-averaged p_T -differential $B^\pm$ production cross section in pp collisions at $\sqrt{s} = 13.6$ TeV and $|y| < 0.8$, compared with the CMS $B^\pm$ measurement at $\sqrt{s} = 13$ TeV [9] and the ALICE B^0 measurement at $\sqrt{s} = 13.6$ TeV [10] (see legend for details). Vertical bars and boxes represent statistical and systematic uncertainties, respectively. Normalization uncertainties are quoted separately.

Candidate J/ψ mesons are reconstructed from oppositely charged electron pairs in events recorded with an online trigger requiring a dielectron candidate. The pairs are required to satisfy a selection on their invariant mass and are then combined with charged-kaon candidates, identified using information from the TPC and TOF, to form $B^\pm$ candidates. Selections on the secondary-vertex fit quality suppress combinatorial background, while a requirement on the decay length of the $B^\pm$ candidate suppresses prompt J/ψ mesons randomly combined with kaon candidates.

The signal yield is extracted from a fit to the invariant-mass distribution of the selected $B^\pm$ candidates. The combinatorial background is described by an analytical function, while the correlated-background shapes and their normalizations relative to the signal are constrained using Monte Carlo simulations, with some variation allowed in the fit. The left panel of Figure 2 shows the fit in the interval $6 < p_T < 8$ GeV/c.

The extracted yields are corrected for the detector acceptance and efficiency and normalized by the branching-ratio product, $\text{BR}(B^\pm \rightarrow J/\psi K^\pm) \times \text{BR}(J/\psi \rightarrow e^+ e^-) = (6.09 \pm 0.12) \times 10^{-5}$, and the integrated luminosity, $\mathcal{L}_{\text{int}} = (49.6 \pm 1.9) \text{ pb}^{-1}$. The resulting p_T -differential production cross section is compared in the right panel of Figure 2 with the $B^\pm$ measurement from CMS [9] and the ALICE B^0 measurement [10]. The measurements show a similar p_T dependence and are generally compatible within uncertainties.

A primary motivation for measuring beauty production in pp collisions is to test perturbative QCD calculations. The left panel of Figure 3 therefore compares the measured p_T -differential $B^\pm$ production cross section with predictions from FONLL [11], GM-VFNS with modified factorization and renormalization scales [12], SACOT- m_T [13], and k_T -factorization [14]. The right panel shows the ratio of the measured cross section to each model prediction. The measured spectrum is broadly described within the theoretical uncertainties. The FONLL and modified-scale GM-VFNS predictions reproduce the experimental values over most of the measured p_T range. The SACOT- m_T prediction tends to lie above the measurement, while the k_T -factorization prediction tends to lie below it, although both remain compatible within uncertainties.

4. Conclusion

The preliminary ALICE measurements presented here demonstrate complementary approaches to beauty production in pp collisions at $\sqrt{s} = 13.6$ TeV. The first preliminary measurement of the non-prompt $\psi(2S)$ fraction provides

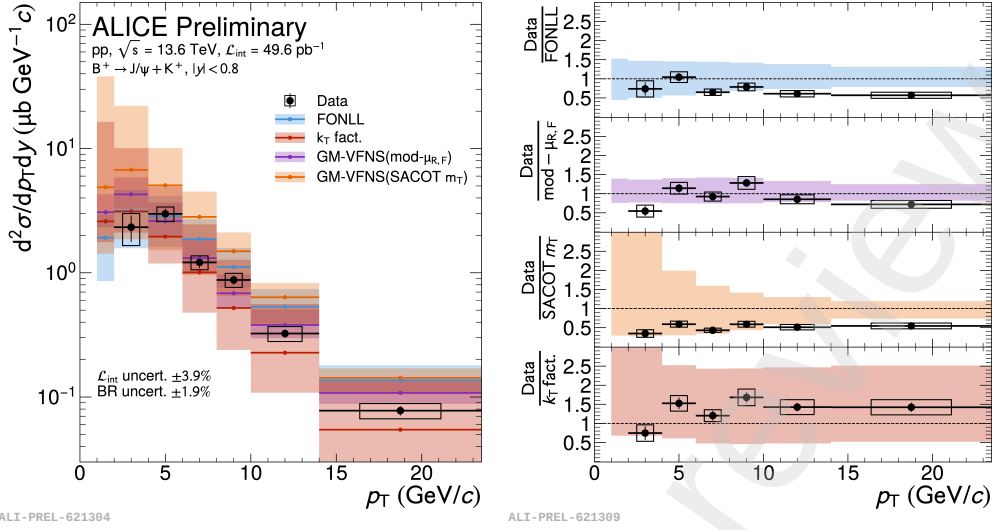


Figure 3: Charge-averaged p_T -differential $B^\pm$ production cross section in pp collisions at $\sqrt{s} = 13.6$ TeV and $|y| < 0.8$, compared with predictions from FONLL [11], modified-scale GM-VFNS [12], SACOT- m_T [13], and k_T -factorization [14] (left), and the corresponding data-to-model ratios (right). Vertical bars and boxes represent statistical and systematic uncertainties, respectively, while the bands show the theoretical uncertainties.

information inclusive over beauty-hadron species, while the exclusive $B^\pm$ measurement provides a species-specific production cross section and extends fully reconstructed measurements at midrapidity towards lower p_T , reaching a previously unexplored kinematic region. The measured $B^\pm$ production cross section is broadly described by state-of-the-art perturbative-QCD calculations. It provides both the pp reference and the reconstruction techniques for future studies of modifications of beauty production in Pb–Pb collisions. In pp collisions, comparisons with measurements of other beauty-hadron species may help constrain beauty-quark hadronization [15]. Together, the $B^\pm$ cross-section measurement presented here and the recently published B^0 cross-section measurement [10] establish a new program of fully reconstructed beauty-hadron measurements in ALICE.

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